

- [CPSC 375: High-Performance Computing](#)

[CPSC 375: High-Performance Computing](#)

Spring 2026

- [Syllabus](#)
- [Schedule](#)
- [Upload](#)
- [The Cluster Project](#)

Homework 15

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. You are designing a parallel system for a weather forecasting model that must simulate atmospheric pressure, temperature, and wind speed across a 3D grid of the Earth. Describe a domain decomposition approach for this problem. How would you divide the data, and what would a primitive task represent?
2. You have a parallel algorithm consisting of 1,000 primitive tasks. You plan to run this on one of the Lenovo P340 SFF compute nodes in your cluster (8 cores, 16 logical processors).
 - A. If you decide to use static load balancing, what is the minimum number of tasks you should aim for to satisfy the “10:1 ratio” rule mentioned in the Mapping Checklist?
 - B. If the communication cost between these 1,000 tasks is high, would you prefer to keep the tasks small or agglomerate them into larger blocks? Explain the trade-off between communication overhead and processor utilization.

• Welcome: Sean

- [LogOut](#)

